

Grapple, Swing, Stabilize: A Prototype Proposal for a Spider-Man–Inspired Robot with Post-Launch Attitude Control

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Abstract— The assignment that motivates this work was stated in two words: build a “Spider-Man robot.” Interpreted literally as a grappling-and-climbing platform, the concept is largely redundant—legged and wheeled climbing robots already scale most surfaces. We argue instead that the defining, biologically-grounded capability of a web-slinging system is not climbing but *dynamic traversal*: rapidly deploying anchors and then controlling the body’s attitude while airborne, so as to swing or jump from anchor to anchor at speed. A survey of tethered perching, surface-attachment taxonomies, and animal anchoring biomechanics converges on a single under-explored problem—*post-launch attitude control*—which most jumping-robot research neglects in favor of take-off performance. The jumping spider (Salticidae) demonstrates a solution: it modulates the tension of an anchored dragline as an internal friction brake to reverse its body pitch and land stably. We formalize the design space along a single rigid-to-soft axis, present seven prototype directions—each with a mechanism diagram—and select a lead direction: a compact, robust, two-wheeled swinger that fires a passive hook and regulates its mid-air attitude by a braked reel and/or a moving-ballast arm. We close with a concrete, low-cost validation experiment modeled directly on the salticid silk-versus-no-silk comparison.

Keywords— grappling robot, tethered locomotion, pendulum dynamics, attitude control, bio-inspired robotics, jumping spider, capstan friction, soft robotics.

1. Introduction

Robotic locomotion over vertical and cluttered three-dimensional environments remains a persistent challenge. Aerial platforms move and see well but are power-hungry, short-endurance, and acoustically conspicuous; ground platforms carry payload and endure but cannot readily gain height. The popular image of Spider-Man suggests a third mode—anchoring a line to the environment and using it to swing or vault rapidly through space—that neither a conventional climber nor a conventional flyer achieves. This proposal takes that image as an engineering brief and asks what, precisely, is worth building.

Our central claim is that the value of a grappling mechanism has been mis-located. If the grapple merely helps a robot ascend a wall, it competes with a mature literature of climbing robots that already handle such tasks with adhesion, magnets, suction, or microspines [4]. What those systems do *not* provide is fast, dynamic, point-to-point traversal exploiting passive pendulum energy. In the Spider-Man analogy, web-slinging is a tool for *motion*, not merely for *attachment*. Reframed this way, the project’s hard problems become (i) deploying an anchor quickly and reliably, and (ii) controlling the robot’s attitude once it has left the ground—the phase during which the center of mass follows an uncontrollable ballistic arc and only body posture remains steerable.

This attitude-control problem is precisely where nature offers guidance and where the robotics literature is thinnest. The jumping spider stabilizes its aerial phase not

with flapping or flailing limbs but by regulating the tension in an anchored dragline [6]. Most spider- and insect-jumping studies, and most jumping robots, concentrate on take-off mechanisms and jump distance; how the body rotates and settles *after* launch is comparatively unaddressed. That gap is the thesis of this proposal. We contribute a literature synthesis, a rigid-to-soft design-space formulation with an anchoring-parameter framework and two locomotion modes, seven prototype directions each with a mechanism diagram, and a selected lead direction with a validation experiment.

2. Background and Related Work

2.1 Tethered perching and grappling

Tethered, or tensile, perching lets an aerial robot conserve energy by hanging from a structure rather than hovering, and can dramatically reduce power consumption for long-duration monitoring [1], [2] (Figs. 1–2). Two mechanical lessons carry into our design. First, the holding force of a line wrapped around a structure grows exponentially with the total wrap angle—the classical capstan relation—so adding wraps is a far stronger lever than counterweight, and it makes attachment nearly independent of surface material [1], [11]. Second, a controlled winding mechanism (spool, servo, helical guide) is essential, or the tether piles unevenly and jams the reel. The physics of ballistically launching a hook that drags a rope behind it is a variable-mass problem that bounds achievable range for a given launch energy [10].

Notably, a mobile robot that launches, anchors, and recovers a detachable hook—explicitly termed a “Spiderman locomotion” system—was demonstrated over a decade ago with a pneumatic launcher and braked reel [9]; launch-and-recover grappling is therefore not itself novel, which sharpens where our contribution must lie.



Fig. 1. An aerial robot perching in a tree by wrapping a tether-borne pod around a branch. From [1], arXiv:2403.01890; © the authors, academic fair use.

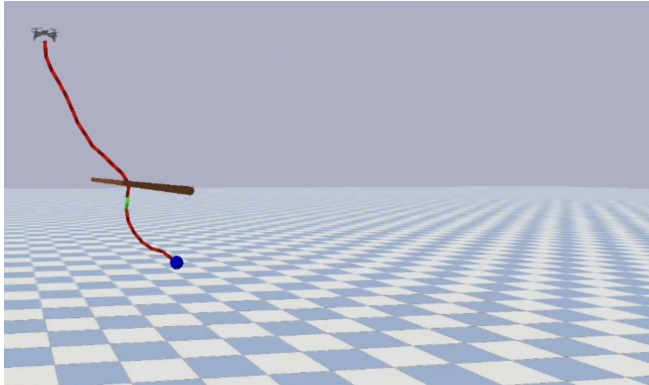


Fig. 2. Simulated two-link robot executing a tensile perch on a bar. From [2], arXiv:2507.06172; © the authors, academic fair use.

2.2 Surface attachment: a taxonomy and its limits

A recent review organizes surface attachment into magnetic adhesion, negative-pressure (suction-cup and suction-chamber) adsorption, bio-inspired dry and wet adhesion, and other principles such as thrust and electrostatic adhesion [4] (Fig. 3). Each is strongly environment-dependent. Magnetic adhesion is mature and strong but needs ferromagnetic surfaces; suction is versatile on smooth surfaces but scales poorly or grows bulky; wet adhesion and electrostatic methods remain weak and precision-sensitive; thrust adhesion is limited by propeller force. Dry adhesion (gecko-inspired) and microspine arrays are most relevant here because they are purely mechanical, reversible, and consumable-free.



Fig. 3. Taxonomy of climbing-robot locomotion mechanisms: continuous (wheeled, crawler), intermittent (legged, inchworm, clamp), and fused. Reproduced from [4], *National Science Review* 2023; CC-BY 4.0.

Microspines—arrays of small, independently sprung hooks that catch surface asperities by mechanical interlocking—recur as a robust, load-scalable strategy. Grippers of hundreds of spines can support an entire robot in any orientation on natural rock, including fully inverted [13] (Figs. 4–5). Their engagement is directional—biting under load one way, releasing the other—so they naturally satisfy a grip-on-load, release-on-command requirement.



Fig. 4. A microspine gripper array interlocking with rough rock. NASA/JPL-Caltech, public domain.



Fig. 5. The LEMUR IIB robot suspended *inverted* from a rock slab on microspine feet [13]. NASA/JPL-Caltech, public domain.

2.3 Biological anchoring and dynamic control

Biology supplies both anchoring and airborne-control strategies. The bagworm builds a ladder-like silk foothold and spreads its weight over roughly twenty glued points, gaining a large safety margin through redundancy rather than any single anchor's strength; its attachment depends strongly on loading direction (shear holds far better than peel) and on substrate roughness and wettability [5]. This argues against consumable adhesive anchoring in an uncontrolled environment and for substrate-agnostic mechanical wrapping.

The most instructive organism is the jumping spider. Chen et al. show its dragline is far more than a safety line: taking off with a rearward pitch, the spider actively adjusts silk tension to reverse its body orientation in the air and land at a shallow, stable angle [6] (Fig. 6). Spiders with silk stay airborne longer, decelerate more, and dissipate about 76% of their mechanical energy versus about 33% without silk, while take-off velocity and jump distance are statistically unchanged—isolating the difference to the dragline. Control is exerted by an internal friction brake (the spinneret valve) clamping the paying-out silk, applied repeatedly through multiple pitch reversals per jump. This is a closed-loop, tension-modulated attitude controller realized with a single tether—a mechanism a braked reel can plausibly emulate.

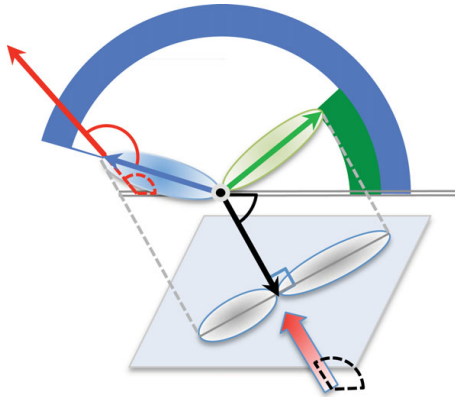


Fig. 6. The jumping spider's in-air reorientation: dragline tension reverses body pitch for a stable landing—the mechanism behind the lead direction. From [6], *J. R. Soc. Interface*; © The Royal Society, academic fair use.

Two further threads bound the design space. Underactuated swinging robots, such as two-link free-flying “trapeze” systems, show how a moving link pumps and steers a swing and how a body that cannot alter its ballistic center-of-mass path can still rotate its posture so a grasping point reaches a target [7]. Separately, soft and continuum robots conform to arbitrary structures—attractive for “grab-anything” anchoring—at the cost of control difficulty and reduced force; recent in-situ web-spinning robots illustrate on-demand fabrication but rely on a consumable material

[8] (Fig. 7). We deliberately exclude brachiation (swinging by grasping pre-existing handholds) because our system must create its own anchors.

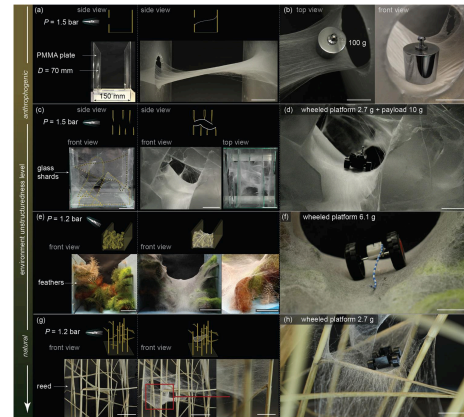


Fig. 7. In-situ spun web structures anchoring across varied substrates and letting a wheeled platform cross debris. From [8], *npj Robotics*; CC-BY-NC-ND 4.0.

3. Problem Formulation and Design Space

3.1 The core capability and its governing constraint

We define the target capability as *rapid, repeatable anchoring followed by controlled aerial traversal*. A single kinematic constraint dominates the airborne phase: once the robot leaves the ground, external forces act only on its center of mass, whose trajectory is a fixed ballistic (or tether-constrained pendular) arc the robot cannot re-aim. What it *can* change is its posture—its orientation about the center of mass. Effective designs therefore do not steer position in flight; they reorient the body so the grasping end reaches the next anchor and the landing is stable. Both the trapeze robot [7] and the jumping spider [6] embody this principle by different means.

3.2 Parameters for a reliable anchor

Reliable anchoring decomposes into: geometric fit (hook gape versus structure diameter, which sets stand-off distance); engagement (whether a catchable feature or sufficient roughness exists, and how many contact points share the load); holding (wrap angle and friction, by the exponential capstan relation); load direction (the shear-versus-peel distinction, favoring geometries that seat under load rather than pry loose); a safety margin above the working load; and, for dynamic use, the probability of catching on a single attempt under impact. This last term is the one slow climbing systems never confront and a fast swinging system lives or dies by.

3.3 The rigid-soft axis and two locomotion modes

Nearly every mechanism choice resolves onto one spectrum. Rigid, mechanical solutions are precise, strong, reusable, and controllable but grip a limited range of surfaces; soft, high-degree-of-freedom solutions conform

to almost anything and tolerate impact but are imprecise, weaker, and often consumable. Because load-bearing and precision favor rigidity while “grab-anything” favors compliance, the likely resolution is a hybrid—a rigid core with a soft conforming tip. Two bio-grounded modes follow: in *swing* mode the robot anchors overhead and travels as a pendulum; in *jump-and-stabilize* mode it makes a ballistic jump with a trailing anchored line and modulates tension to reverse pitch for a controlled landing, after the salticid [6].

These modes are best composed rather than opposed, because an initial jump also serves a dynamic purpose at every swing. Entering a taut-tether swing from rest forces the tether to bend a stationary body abruptly onto a circular arc—a near-impulsive tension that spikes both the peak force and the acceleration the structure and payload must survive. Arriving with velocity already matched to the arc reduces this to a smooth centripetal force. In arboreal-locomotion mechanics this flight-to-swing transition is treated explicitly as a *collision* whose severity a velocity match minimizes and a mismatch maximizes [15], [16]. The practical consequence is that a small jump should *prime* the entry into swinging—jump, match, swing—across all variants, lowering peak tether load, body acceleration, and actuator torque at engagement. (We cite this brachiation result for its transition dynamics only; our system creates its own anchors rather than grasping fixed handholds.)

4. Proposed Prototype Directions

The seven directions below span the rigid-to-soft axis and are a menu of options rather than competing final designs. Each is stated as a mechanism (how it is assumed to work), a function (what it delivers), and an assessment (its principal open problem), with a schematic.

4.1 P1 — Smart grappling hook

Mechanism. A wheeled ground unit carries the battery, computation, and a braked reel; the tether doubles as a conductor powering a small single-rotor drone, which flies to a branch and coils the tether around it as a capstan wrap. The ground unit sits in an actuated gimbal ring that keeps it level while winching and rotates to present it to a surface. **Function.** This converts a thrown, uncontrollable hook into a precisely placed one, opening many more mounting points while heavy sensing stays on the ground unit. **Assessment.** The drone is acoustically conspicuous, and a tether carrying both current and structural load is a demanding cable whose resistance grows with length, trading reach against power. P1 is the origin sketch that motivates P5 and P6.

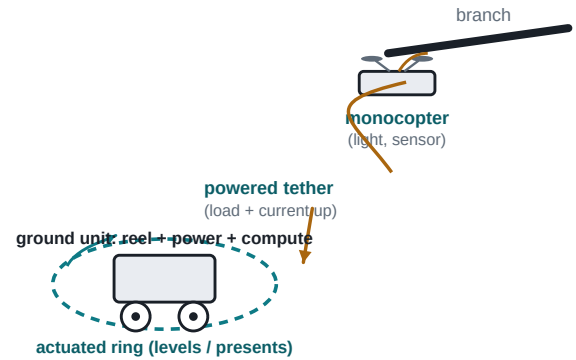


Fig. 8. P1: a light monocopter places and wraps a powered tether around a branch; the ground unit rides in an actuated ring that levels it and rotates it to meet a surface.

4.2 P2 — Ladder / dual-attachment

Mechanism. A deployable ladder-like double-rail anchors at two points, its rungs distributing load across many contacts—mirroring the bagworm foothold [5]. **Function.** Its advantage is stability—resisting sway and rotation, and spanning a gap—rather than raw strength, since a single wrapped point already holds exponentially well. **Assessment.** The central open problem is autonomous connection of the two ends; the span is mechanical, not a spun or glued web.

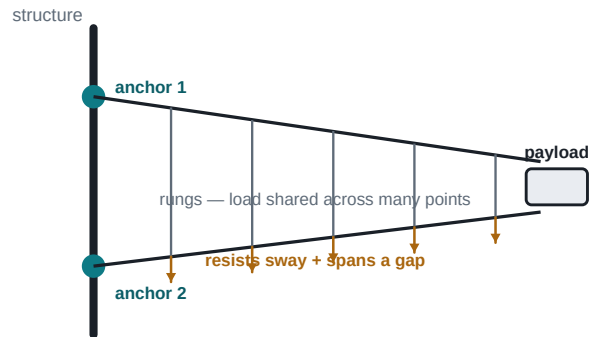


Fig. 9. P2: two anchor points carry a ladder-like span whose rungs share the payload load; the value is anti-rotation stability and gap-spanning, not peak strength.

4.3 P3 — Vacuum grapppler

Mechanism. A fan-evacuated suction chamber—area far larger than a cup’s, giving strong, leak-tolerant negative pressure—clamps onto smooth, featureless surfaces. **Function.** It covers exactly the case a wrap or hook cannot: flat walls, glass, panels. Its failure surface (rough, porous) is the complement of the hook’s, so the two pair naturally. **Assessment.** Cup arrays scale badly and fan chambers are bulky and loud, so P3 is a niche add-on; a flyer’s rotors could serve as the suction source (power reuse).

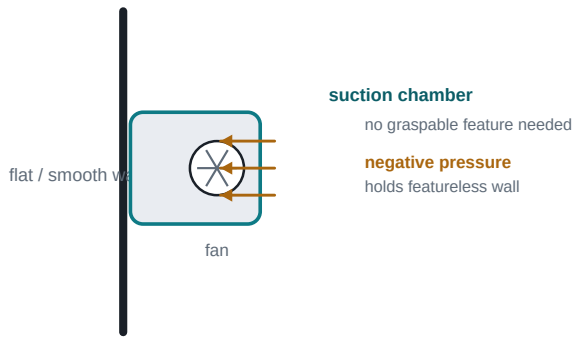


Fig. 10. P3: a fan evacuates a chamber to hold a featureless wall by negative pressure—covering the smooth surfaces on which a hook or wrap has nothing to grip.

4.4 P4 — Hybrid grapple-and-climber

Mechanism. One platform senses the surface and selects a mode: grapple-and-scale on a wrappable feature, or on-board climbing (microspines or suction) on a bare wall. **Function.** Together the modes approach complete coverage—grapple gives fast vertical gain, climbing handles flat walls. **Assessment.** The cost is a genuine multimodal-locomotion problem—surface sensing, a mode policy, and two subsystems’ mass—and it re-inherits the redundancy critique unless grapple stays dominant.

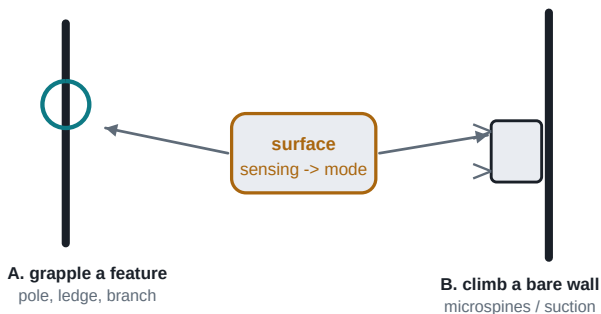


Fig. 11. P4: surface sensing selects between grapppling a feature (mode A) and climbing a bare wall with microspines or suction (mode B).

4.5 P5 — Drone as anchor

Mechanism. Inverting P1, the drone is the primary agent and a self-placing flying anchor, tethered to a simple, heavy ground unit holding power, computation, and the winch (Fig. 12). **Function.** Because the drone *defines* the anchor, the system can swing where no natural anchor exists, and the separated bodies give a wide baseline for two-viewpoint sensor fusion and improved 3-D anchor detection; winching against a wrapped anchor is far cheaper than flying the mass up a steep slope, and a battery-swap dock can extend endurance. **Assessment.** The tether’s role must be pinned down, since load-bearing, sensing, and power imply different cables;

the two-agent architecture has a strong secondary application to defense and field robotics [3].

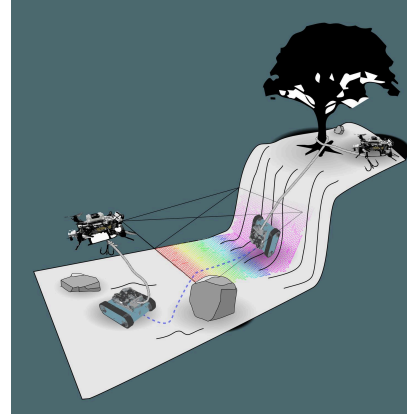


Fig. 12. Concept of a UAV assisting a UGV to ascend a cliff by placing an anchored tether—the direct inspiration for P5. From [3], arXiv:1903.04898; © the authors, academic fair use.

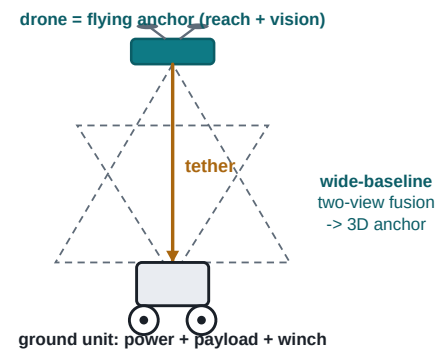


Fig. 13. P5: the drone is a self-placing flying anchor over a simple ground unit; their separated viewpoints give a wide baseline for two-view sensor fusion and 3-D anchor detection.

4.6 P6 — Two-wheel swinger with inertia arm (lead direction)

Mechanism. A small, deliberately rugged two-wheeled body carries a central launcher for a passive, recoverable hook and a ballast mass on a moving arm. It fires the hook, swings as a pendulum, and shifts the ballast to change its moment of inertia during the swing—pumping, steering, and timing the release. Following the trapeze principle [7], the arm’s true function is posture control to place the grasping end, not to move the center of mass. A second, spider-inspired option is a braked reel that modulates tether tension to reverse pitch [6]; the two may be combined. **Function.** P6 is the simplest embodiment that exercises the full thesis, and its minimal, rugged form prioritizes durability. Consistent with Section 3.3, it initiates each swing with a small jump so its velocity matches the tether arc, avoiding the near-impulsive load of a from-rest engagement and cutting the peak tension, acceleration, and torque at catch [15]. **Assessment.** Open problems are self-righting and steering between swings, the hook type that maximizes surface breadth (motivating

a soft tip, P7), single-attempt catch reliability under impact, and the ballast arm's control authority (its mass times arm length against the body mass). This is the lead direction, developed in Section 5.

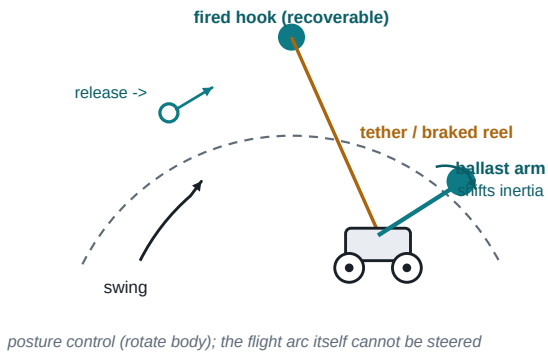


Fig. 14. P6 (lead): the robot fires a recoverable hook, swings as a pendulum, and shifts a ballast arm (or brakes the reel) to rotate its body and place the grasping end before releasing. The flight arc itself cannot be steered—only posture.

4.7 P7 — Soft-body active grapple

Mechanism. A continuum, octopus-arm-like limb reaches out and coils around a structure, conforming to almost any shape and forming multiple capstan wraps without a catchable edge; candidate actuation includes tendon drives, pneumatic muscles, or shape-memory alloy. **Function.** It attacks the surface-breadth limit of a rigid hook while staying mechanical, reusable, and impact-tolerant. **Assessment.** Unbounded degrees of freedom bring hard control, lower force, and slower actuation, poorly suited to load-bearing and fast catching; its best role is a soft conforming tip on the rigid core of P6—a hybrid that buys breadth without sacrificing toughness.

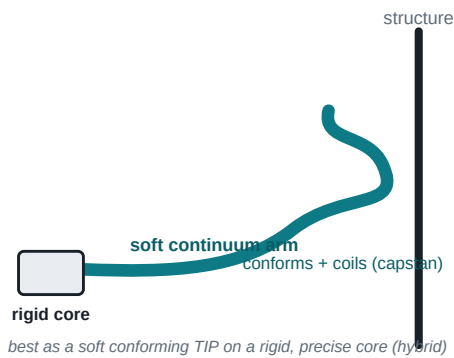


Fig. 15. P7: a soft continuum arm on a rigid core reaches out and coils a structure, conforming where a rigid hook cannot; best deployed as the soft tip of a rigid-core hybrid.

5. Selected Direction and Preliminary Plan

We propose to develop P6, the two-wheel swinger, as the lead platform: it is the minimal design that deploys its own anchor, traverses by swinging, and must control attitude in flight, and its rugged base occupies the

well-understood wheeled region of Fig. 3.

The scientific core is a direct comparison of two attitude-control mechanisms for the airborne phase. The first is the moving-ballast arm, which changes the body's moment of inertia, after the underactuated trapeze approach [7]. The second, from the jumping spider [6], is a braked reel that modulates tether tension to brake the body and reverse its pitch; it is likely lighter and reuses the reel the system already needs for recovery. The two may be combined.

We propose to evaluate them with an experiment modeled directly on the salticid silk-versus-no-silk comparison [6]. The platform performs repeated jumps or swings under three conditions—no active control, braked-reel control, and ballast-arm control—while high-speed video records the same variables measured for the spider: aerial-phase duration, horizontal deceleration, energy dissipation, and body pitch angle at landing. Success is reproducing, under active control, the spider's demonstrated reduction of landing pitch and increase of energy dissipation relative to the uncontrolled baseline. Such a result would demonstrate the mechanism cleanly, yield a quantitative controller specification, and constitute a self-contained, publishable contribution.

6. Discussion and Open Questions

Several decisions remain open and frame the first design phase. (Q1) Should swing or jump-and-stabilize be primary, or should they be sequenced? (Q2) Ballast arm, braked reel, or both—resolved by the experiment of Section 5? (Q3) Where on the rigid-soft axis does the end-effector belong; does a soft conforming tip earn its control cost? (Q4) What is the tether's primary function, given that load-bearing, sensing, and power imply different cables? (Q5) Should any consumable web-material capability be admitted under the reusability constraint, and only if a recoverable fiber proves feasible? On novelty: launch-and-recover grapppling robots exist [9], and swinging acrobatic robots exist [7]; the contribution is their union under a specific, under-studied objective—post-launch attitude control for a self-anchoring dynamic traverser—grounded in a concrete animal mechanism. That framing also makes the project appropriate to a biorobotics setting: the differentiator is a biological control strategy, not a new actuator.

7. Conclusion

A “Spider-Man robot” is most interesting not as a climber but as a self-anchoring dynamic traverser, and its hardest, most novel problem is controlling body attitude after launch—which the jumping spider solves with a

tension-modulated dragline. From a survey spanning tethered perching, surface attachment, and animal anchoring, we distilled a rigid-to-soft design space, an anchoring-parameter framework, and two locomotion modes, and presented seven prototype directions with mechanism diagrams. We recommend a compact, rugged two-wheel swinger as the lead platform and a salticid-inspired experiment comparing braked-reel and ballast-arm attitude control as the first concrete step.

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